MSCF Probability, Summer 2018: Homework #3 Solutions

1. **3 points**

   The function $g(x) = 1/x$ is such that $g''(x) = 2/x^3 > 0$ for $x > 0$. Hence, this function is convex on the support of X. Conclude from Jensen's inequality that $E(1/X) \geq 1/E(X)$.

2. **7 points: (a) 3 pts (b) 2 pts (c) 2 pts**

   (a)

$$
\begin{aligned}
f_X(x) &= \int_y f_{X,Y}(x,y)dy = \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} \frac{1}{\pi} dy \\
&= \begin{cases} \frac{2}{\pi}\sqrt{1-x^2} & x^2 \leq 1 \\ 0 & \text{otherwise} \end{cases}
\end{aligned}
$$

By symmetry, the marginal density of Y is;

$$
f_Y(y) = \begin{cases} \frac{2}{\pi}\sqrt{1-y^2} & y^2 \leq 1 \\ 0 & \text{otherwise} \end{cases}
$$

   (b)

$$
\begin{aligned}
F_D(d) &= P(\sqrt{X^2+Y^2} \leq d) = P(X^2+Y^2 \leq d^2) \\
&= \underbrace{\int\int}_{x^2+y^2 \leq d^2} \frac{1}{\pi} dxdy = \frac{1}{\pi} \underbrace{\int\int}_{x^2+y^2 \leq d^2} dxdy
\end{aligned}
$$

Note that the double integral is the area of a circle of radius d and thus equal to πd^2

$$
\Rightarrow F_D(d) = \frac{1}{\pi} \cdot \pi d^2 = \begin{cases} 0 & d \leq 0 \\ d^2 & 0 < d < 1 \\ 1 & d > 1 \end{cases}
$$

   (c)

$$
\begin{aligned}
f_D(d) &= \frac{\partial}{\partial d} F_D(d) = 2d,\ 0 < d < 1 \\
&\Rightarrow\ E(D) = \int_{d=0}^{1} d \cdot 2d\, dd = \frac{2d^3}{3}\Big|_0^1 = \frac{2}{3}
\end{aligned}
$$

3.

$$
\begin{aligned}
F_T(t) &= P(T \leq t) = P(XY \leq t) = P(Y < t/X) \\
&= \int_{y=0}^{\infty} \int_{x=0}^{t/y} y e^{-y(x+1)} dx dy \\
&= \int_{y=0}^{\infty} e^{-y} \int_{x=0}^{t/y} y e^{-yx} dx dy \\
&= \int_{y=0}^{\infty} e^{-y} \cdot (1 - e^{-y \cdot t/y})\, dy \\
&= ((1 - e^{-t}) \int_{y=0}^{\infty} e^{-y} dy = 1 - e^{-t} \quad , t > 0 \\
&\Rightarrow \; f_T(t) = \frac{\partial}{\partial t} F_T(t) = e^{-t} \quad t > 0 \quad \Longrightarrow \quad T \sim Exp(1)
\end{aligned}
$$

4. (a) In lecture, we established that the joint p.d.f. of $U_{(1)}$ and $U_{(2)}$ is given by

$$
f_{U_{(1)},U_{(2)}}(u_1, u_2) = \begin{cases} 2 & \text{if } 0 \leq u_1 \leq u_2 \leq 1 \\ 0 & \text{otherwise} \end{cases}
$$

Now, we defined L_1 and L_2 by $L_1 = U_{(1)}$ and $L_2 = U_{(2)} - U_{(1)}$. So $L_1 + L_2 = U_{(2)}$. We know that the support for (L_1, L_2) will be any values of ℓ_1 and ℓ_2 such that $0 \leq \ell_1 \leq \ell_1 + \ell_2 \leq 1$. For these choices of ℓ_1 and ℓ_2,

$$
\begin{aligned}
P(L_1 \leq \ell_1, L_2 \leq \ell_2) &= P(U_{(1)} \leq \ell_1, U_{(2)} - U_{(1)} \leq \ell_2) \\
&= P(U_{(2)} - \ell_2 \leq U_{(1)} \leq \ell_1) \qquad (1)
\end{aligned}
$$

Since we know that the joint density of $U_{(1)}$ and $U_{(2)}$ is uniform (i.e. $= 2$) on its support, calculating the probability in (1) will just amount to computing twice the area of the region where the event in (1) holds. The figure above gives the relevant region, a parallelogram with base ℓ_2 and height ℓ_1; thus, when $0 \leq \ell_1 \leq \ell_1 + \ell_2 \leq 1$,

$$
F_{L_1,L_2}(\ell_1, \ell_2) = P(L_1 \leq \ell_1, L_2 \leq \ell_2) \;=\; 2\ell_1\ell_2
$$

To find the joint p.d.f. $f_{L_1,L_2}(\ell_1, \ell_2)$, we take $\frac{\partial^2 F_{L_1,L_2}}{\partial \ell_1 \ell_2}$, which yields:

$$
f_{L_1,L_2}(\ell_1, \ell_2) = \begin{cases} 2 & \text{if } 0 \leq \ell_1 \leq \ell_1 + \ell_2 \leq 1 \\ 0 & \text{otherwise} \end{cases}
$$

(b) Then, to find $f_{L_1}(\ell_1)$ and $f_{L_2}(\ell_2)$, we integrate the joint distribution over the other variable in that region. For $0 \leq \ell_1 \leq 1$,

$$
f_{L_1}(\ell_1) = \int_0^{1-\ell_1} f_{L_1,L_2}(\ell_1, \ell_2) d\ell_2 = \int_0^{1-\ell_1} 2 d\ell_2 = 2 - 2\ell_1
$$

Similarly, for $0 \leq \ell_2 \leq 1$,

$$f_{L_2}(\ell_2) = \int_0^{1-\ell_2} f_{L_1,L_2}(\ell_1, \ell_2) d\ell_1 = \int_0^{1-\ell_2} 2 d\ell_1 = 2 - 2\ell_2.$$

Finally, note that the $Beta(1, 2)$ distribution has density

$$\frac{\Gamma(3)}{\Gamma(1)\Gamma(2)} t^{1-1}(1-t)^{2-1} = 2 \cdot (1-t) = 2 - 2t.$$

restricted to the interval t $\in$ [0,1].
Hence, L_1 and L_2 both have a $Beta(1, 2)$ distribution.